

EEE 5354L:
Semiconductor Device Fabrication Laboratory
Device Fabrication SOP
Version 1.3.1

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Lab 1: Thermal Oxidation (Field Oxide)

Prepare the Tube Furnace

1. Make sure the oxidation tube is installed in the furnace
2. Open both valves on the Oxygen tank
3. Replace the nitrogen line on the end of the tube furnace with the line attached to the bubbler
4. Plug in and turn on the heat gun, and set it up underneath the end of the tube furnace
5. Turn on the tube furnace and set it to **1000 °C**
6. Turn on the bubbler hotplate and line up the dial with the Sharpie mark
7. Make sure the gas flow value is set to “Wet Oxidation”
8. Set the oxygen flow for **1 L/min**
9. While the tube furnace and hot plate are heating, prepare the wafers

Prepare the wafers

1. Get the low-resistivity (1-10 Ωcm) P-type wafers from the class drawer in the teaching lab
2. Use a diamond scribe to mark the back of the wafers with your initials
3. Rinse the wafer with DI water to get rid of the dust created after scribing and dry them using the nitrogen blower
4. Place the wafers into the oxidation boat
 - The flats should be facing up
 - This makes them easier to grab
5. Place two extra “dummy” wafers in front and behind your “process” wafers
6. Wait until the furnace and hotplate reach temperature

Oxidation

1. Make sure the water in the bubbler isn't boiling
 - The water in the bubbler should be hot, but not boiling
 - If the water starts boiling, turn down the hotplate slightly
 - The temperature of the bubbler should be around 90 °C
2. There might be a little water pooling at the end of the furnace
 - This is normal
 - Make sure the heat gun is blowing on this area
3. Put on heat-resistant gloves and remove the glass cap from the furnace
4. Use the metal tongs and sheet to put the boat into the tube furnace
5. Use the glass rod to load the wafers into the furnace
 - **MAKE SURE YOU DO THIS VERY SLOWLY OR THE WAFERS AND/OR BOAT WILL CRACK DUE TO THERMAL STRESS**
 - You should move about 1 in every 10 seconds
 - There should be a Sharpie mark on the quartz rod
 - The wafers will be centered in the tube once this mark lines up with the edge of the tube
 - Take care not to bang the rod against the furnace, as this could cause the rod to shatter
 - **THE ROD ACTS AS A WAVEGUIDE FOR IR ENERGY. THIS MEANS THE END OF THE ROD WILL REACH SIMILAR TEMPERATURES TO THE INSIDE OF THE TUBE FURNACE, SO MAKE SURE TO KEEP IT AWAY FROM YOUR BODY AND OTHER PEOPLE**
6. Once the wafers are loaded, put the glass end cap back on the tube
7. Let the wafers oxidize for **80 minutes**
 - This should grow about 400 nm of thermal oxide
8. After 65 minutes, use the glass rod to unload the wafers from the furnace
 - Use the same rate of 1 cm every 10 seconds to prevent thermal shock

9. Use the filmmetric tool to measure the oxide thickness
 - Make sure the light source is on for at least 5 minutes so the light source can stabilize
10. If the oxide thickness is under 400 nm, use the [BYU oxidation calculator](#) to find out how much longer you need to oxidize
 - It is fine if the thickness is **above 400 nm**
11. Once the desired oxide thickness is achieved, make sure to clean up the teaching lab

Clean up the Teaching Lab

1. Clean up the teaching lab
2. Make sure the furnace is turned off
3. Make sure the oxygen flow rate is set to 0 L/min
4. Make sure to turn the gas valve to “Off”
5. Make sure both valves on the oxygen tank are turned off
 - **Do not over-tighten the valves**
6. Make sure the hot plate is turned off
7. replace the bubbler line with the main nitrogen line
8. The furnace will still be hot
9. Make sure you do not touch it
10. Turn off and unplug the heat gun

Lab 2: Mask #1 Lithography: Diffused Regions and BOE Etch

Lithography

Prepare the Wafers

1. Turn on the photolithography hotplate
2. Wait for the hotplate to reach **112 °C**
 - The hotplate will overshoot before stabilizing at 112 °C
3. Place the wafers on the hot plate and let them bake for **10 minutes**
 - This is the dehydration bake

Prepare the Spinner

1. While the wafers are dehydrating, set up the spinner — using the **Lithography SOP**
 - (a) Fold a piece of foil into a square and tear off the corner where the folds meet
 - (b) Unfold the foil and fold it onto the spinner.
 - (c) located the 2 in-chuck and place it onto the spinner
 - (d) turn on the vacuum pump located underneath the left cabinet, underneath the spinner area
 - (e) turn on the spinner located to the left of the hotplate
2. **MAKE SURE THE OVERHEAD LIGHTS ARE TURNED OFF BEFORE PROCEEDING TO THE NEXT STEP**

Photoresist Spinning

1. Once the wafers have finished dehydrating, locate the bottle of **AZ 1512** photoresist and place it next to the spinner
 - This is a **positive** photoresist
2. Center your wafer on the chuck and set your spin parameters on the spinner
 - Speed: **1800 RPM**
 - Time: **30 seconds**

3. use the nitrogen blower to blow off any large particles or dust from the surface of your wafer
4. Dispense about 1.5 mL of photoresist onto the center of your wafer, then press the foot pedal to start spinning
 - This does not need to be exact
5. Once the spinner has stopped, place your wafer onto the hotplate for **2 minutes**
6. Remove the wafer from the hotplate and put it back into the acrylic case

Prepare the Mask Aligner

1. Turn on the light located above the mask aligner
2. turn on the power strip located to the right of the mask aligner
3. Turn on the vacuum pump located on the floor to the right of the mask aligner
4. Make sure the mask aligner is turned on
5. locate **Mask #1: Active Areas** in the TA drawer
6. Place the mask onto the mask holder and turn on the vacuum switch located on the mask aligner
 - (a) There is a specific orientation the mask should be in before placing it onto the holder
 - (b) The bottom of the pattern should be facing up
 - (c) Ask your TA to show you how to identify the correct orientation
7. Verify the mask is secured by flipping it upside down with your hand underneath it
8. Once you confirm the mask is secure, load the mask into the mask aligner
9. Next, perform a test exposure to find the lamp intensity
10. Calculate the exposure time using the current lamp intensity and a dose of **110 $\frac{mJ}{cm^2}$**
 - The equation for exposure time is $time\ (seconds) = \frac{Dose\ (mJ/cm^2)}{Lamp\ Intensity\ (mW/cm^2)}$
 - If your lamp intensity is $8.7\ \frac{mJ}{cm^2}$, your exposure time would be
$$time = \frac{110\ mJ/cm^2}{8.7\ mW/cm^2} = 12.6\ seconds$$

Exposure

1. Once you have your exposure time, make sure you turn the dial to correspond with your exposure time
2. Load your wafer into the mask aligner
3. Make sure your wafer is centered in the chuck
4. Bring your wafer in contact with the mask and press the green “Exposure” button
5. After exposure, unload your wafer and bring it to the solvent fume hood

Development

1. Fill a small glass dish with about 50 mL of **AZ 300 MIF Developer**
2. Place your exposed wafer into the developer for **75 seconds**
 - Make sure to agitate the wafer by stirring the glass dish
3. Rinse the wafer using the DI water faucet for a couple of seconds
 - Make sure to grab a different end of the wafer with the tweezers while rinsing
 - There can be developer underneath the tweezers
4. Dry the wafer using the N₂ gun
 - Make sure to dry underneath the tweezers

Inspection

1. Use the microscope to inspect the patterns and test structures on your wafer
 - Look at the resolution structures and record what features survived

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room

- (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
- (c) Wait for the vacuum gauge to reach 300 mtorr
- (d) Turn on the power strip between the gas flow controller and the RIE tool
- (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
- (f) Turn on the main flow and the flow for both O₂ and Ar
- (g) Press the gas flow 1 button on the RIE
- (h) Wait for the vacuum gauge to stabilize
- (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W
- (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**
- (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
- (l) Press the gas flow 1 button on the RIE to turn off the gas flow
- (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
- (n) Turn off the roughening pump underneath the table
- (o) Press the vent button and wait for the chamber to return to atmosphere
- (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
- (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Optional Step

1. Place the wafer back on the hotplate for 2 minutes
 - This is the “Hard Bake”
 - This will strengthen the resist to survive etch chemistry in later steps

Buffered Oxide Etchant (BOE) Etch

Hydrofluoric Acid (HF) Safety

- BOE contains **HF** and it can be **extremely dangerous**
- An exposure area equivalent to the area of your hand can be fatal
 - Attacks calcium in your body
 - Bones and neurons contain calcium
- Make sure your gloves and apron stay dry at all times
- If you see any clear liquid on your PPE, make sure to rinse and dry it before continuing
- Any spills should be reported to your TA
- Make sure to know the location of the HF response kit before proceeding.

Prepare Acid/Base Fume Hood

1. Locate the following items and place them in the acid/base fume hood
 - (a) Two large **teflon** beakers
 - (b) Two small **teflon** beakers
 - (c) One 2 in wafer holder
 - This will be made from an old bottle
2. Fill the large teflon beakers with DI water and place them in the back of the fumehood
3. Determine the etch time
 - The etch rate of thermal oxide in BOE is about **2 nm/sec (120 nm/min)**
 - Divide the thickness of your oxide by 100 to determine your etch time
 - Example: 400 nm of oxide will result in a **3-minute, 33-second etch**
 - It is also a good practice to etch for a few seconds longer to make sure all the oxide is gone in the desired areas

Put on PPE

1. You should already have on the standard lab PPE

- Lab coat
- safety glasses
- Gloves
- long pants
- closed-toe shoes

2. Put on the acid/base PPE

- Blue acid apron
- faceshield
- Black chemical gloves

BOE Etch

1. Once PPE is on, locate the BOE waste bottle and verify it is not full
2. Place it in the fume hood
3. Take out the fresh BOE and pour it into one of the small teflon beakers
4. Put away fresh BOE when finished
5. Place the wafer into the wafer holder
6. Place wafer into the BOE etchant for the previously determined time
7. Once wafers have finished etching, take the wafer out of the BOE etchant
 - make sure to allow any etchant on the wafer and in the holder to fall back into the BOE without splashing
8. Place the wafer and holder into the first large beaker of DI water for a few seconds
 - Try to prevent water from splashing or dripping onto the edge of the beaker or inside the fumehood
9. Remove the first beaker and place the wafer into the second beaker of DI water
10. Remove the wafer from the second beaker and place it into the empty small teflon beaker

- Make sure any excess water falls back into the water beaker
11. Take the wafer to the DI water faucet and triple rinse the wafer by filling the beaker with water and emptying it.
 12. Once finished, place the wafer to the side and rinse and dry your gloves
 13. Use the N₂ gun to dry your wafer
 14. Use the filmetrics tool to verify the oxide has been completely etched away
 15. Put the wafer back into the BOE if oxide still remains
 16. Once finished etching, clean up the acid fume hood
 - Pour BOE into the waste container
 - Pour rinse water into the acid/base sink
 - Use the tap water faucet to rinse out all the beakers and wafer holders used
 - Put rinsed tools on the drying rack.
 - Rinse and dry the gloves and put away acid/base PPE

Resist Strip

1. Strip resist using acetone or MLO-07 resist strip
2. Using acetone:
 - (a) Pour acetone into a small glass bowl
 - (b) Place wafer into the beaker and agitate until all the resist has been removed
 - (c) Remove wafer and rinse using DI water faucet and dry using N₂ gun
 - (d) Pour any leftover acetone into the acetone waste container
 - (e) Rinse any dishes used and put them on the drying rack
3. Using MLO-07 resist strip:
 - (a) Pour the MLO-07 into a small bowl and place it on the hot plate in the solvent fume hood
 - This hotplate does **NOT** have a wafer on it
 - (b) Set the hotplate to **75 °C**
 - (c) Use a thermometer to verify the temperature of the solution

- (d) Once the temperature has reached at least 60 °C, place the wafers into the solution and agitate until all the resist has been removed
- (e) Turn off the hotplate and place the glass bowl inside the fume hood
- (f) Remove the wafer and rinse using the DI water faucet and dry with the N₂ gun
- (g) Pour any leftover MLO-07 into the MLO-07 waste container
- (h) Rinse any dishes used and put them on the drying rack

Lab 3: Spin-on-Glass (SOG) and Thermal Diffusion

Prepare Tube Furnace

1. Make sure the SOG/Diffusion tube is installed in the tube furnace
2. Make sure the gas flow valve is set to “Dry Oxidation”
 - We are not flowing oxygen during this step, so there should be no oxidation
 - The tube furnace should be connected to the house nitrogen line
3. Set the nitrogen flow for **1 L/min**
4. Turn on the furnace and set it to **1000 °C**
5. You can do the next few steps while the furnace is heating up

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room
 - (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
 - (c) Wait for the vacuum gauge to reach 300 mtorr
 - (d) Turn on the power strip between the gas flow controller and the RIE tool
 - (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
 - (f) Turn on the main flow and the flow for both O₂ and Ar
 - (g) Press the gas flow 1 button on the RIE
 - (h) Wait for the vacuum gauge to stabilize
 - (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W

- (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**
- (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
- (l) Press the gas flow 1 button on the RIE to turn off the gas flow
- (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
- (n) Turn off the roughening pump underneath the table
- (o) Press the vent button and wait for the chamber to return to atmosphere
- (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
- (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Prepare the Spinner and SOG Hotplate

1. Locate the hot plate in the far right-hand corner of the solvent fumehood
 - It should have a large wafer on top of it
2. Turn it on and set it to **175 °C**
3. While the hot plate is warming up, prepare the spinner
4. Fold a piece of foil into a square and tear off the corner where the folds meet
5. Unfold the foil and fold it onto the spinner.
6. Locate the 2 in-chuck and place it onto the spinner
7. Turn on the vacuum pump located underneath the left cabinet, underneath the spinner area
8. Turn on the spinner located to the left of the hotplate

Spin Coat SOG

1. Locate the small bottle of SOG and place it next to the spinner
2. Place the wafer onto the chuck and verify it is centered
3. Use the following parameters to program the spin coater
 - Speed: **3500 RPM**

- Time: **30 seconds**
4. Dispense about 1.5 mL of SOG onto your wafer and press the foot pedal to start the spinner
 5. Once the spinner stops, place the wafer onto the 175 °C hot plate in the solvent fumehood for **10 minutes**
 6. After 10 minutes, remove the wafer from the hot plate and place it in the fume hood for a few seconds
 7. Place the wafer into the acrylic case and bring it to the tube furnace

Thermal Diffusion

1. Place the wafers into the quartz boat meant for thermal diffusion
 - The boat should have a large gash in the middle of it
2. Once all the wafers have been loaded, place a wafer on the front and back ends of the boat
 - These wafers should sandwich your process wafers
3. **PUT ON THE HEAT-RESISTANT GLOVES BEFORE PROCEEDING**
4. Remove the glass end cap from the tube furnace
5. Using the metal plate and the tongs, place the boat holding the wafers into the turn furnace
6. Once the wafers are placed, use the quartz rod to push the wafers into the center of the tube
 - **MAKE SURE YOU DO THIS VERY SLOWLY OR THE WAFERS AND/OR BOAT WILL CRACK DUE TO THERMAL STRESS**
 - You should move about 1 in every 10 seconds
 - There should be a Sharpie mark on the quartz rod
 - The wafers will be centered in the tube once this mark lines up with the edge of the tube
 - Take care to not bang the rod against the furnace, as this could cause the rod to shatter

- **THE END OF THE ROD WILL REACH THE SAME TEMPERATURE AS THE FURNACE, SO MAKE SURE TO KEEP IT AWAY FROM YOUR BODY AND OTHER PEOPLE**

7. Once the wafers have been centered in the tube, remove the quartz rod
8. Let the wafers diffuse for **2 hours**
9. After 2 hours, remove the wafers from the tube furnace following all rules described during loading
10. remove the wafers from the boat and place them into your acrylic box

- **THE BOAT WILL STILL BE EXTREMELY HOT, SO BE CAREFUL WHEN RETRIEVING THE WAFERS**

Lab 4: Mask #2 Lithography: Gate Oxide and BOE Etch

Lithography

Prepare the Wafers

1. Turn on the photolithography hotplate
2. Wait for the hotplate to reach **112 °C**
 - The hotplate will overshoot before stabilizing at 112 °C
3. Place the wafers on the hot plate and let them bake for **10 minutes**
 - This is the dehydration bake

Prepare the Spinner

1. While the wafers are dehydrating, set up the spinner — using the **Lithography SOP**
 - (a) Fold a piece of foil into a square and tear off the corner where the folds meet
 - (b) Unfold the foil and fold it onto the spinner.
 - (c) located the 2 in-chuck and place it onto the spinner
 - (d) turn on the vacuum pump located underneath the left cabinet, underneath the spinner area
 - (e) turn on the spinner located to the left of the hotplate
2. **MAKE SURE THE OVERHEAD LIGHTS ARE TURNED OFF BEFORE PROCEEDING TO THE NEXT STEP**

Photoresist Spinning

1. Once the wafers have finished dehydrating, locate the bottle of **AZ 1512** photoresist and place it next to the spinner
 - This is a **positive** photoresist
2. Center your wafer on the chuck and set your spin parameters on the spinner
 - Speed: **1800 RPM**
 - Time: **30 seconds**

3. use the nitrogen blower to blow off any large particles or dust from the surface of your wafer
4. Dispense about 1.5 mL of photoresist onto the center of your wafer, then press the foot pedal to start spinning
 - This does not need to be exact
5. Once the spinner has stopped, place your wafer onto the hotplate for **2 minutes**
6. Remove the wafer from the hotplate and put it back into the acrylic case

Prepare the Mask Aligner

1. Turn on the light located above the mask aligner
2. turn on the power strip located to the right of the mask aligner
3. Turn on the vacuum pump located on the floor to the right of the mask aligner
4. Make sure the mask aligner is turned on
5. locate **Mask #2: Gate Oxide** in the TA drawer
6. Place the mask onto the mask holder and turn on the vacuum switch located on the mask aligner
 - (a) There is a specific orientation the mask should be in before placing it onto the holder
 - (b) The bottom of the pattern should be facing up
 - (c) Ask your TA to show you how to identify the correct orientation
7. Verify the mask is secured by flipping it upside down with your hand underneath it
8. Once you confirm the mask is secure, load the mask into the mask aligner
9. Next, perform a test exposure to find the lamp intensity
10. Calculate the exposure time using the current lamp intensity and a dose of **110 $\frac{mJ}{cm^2}$**
 - The equation for exposure time is $time\ (seconds) = \frac{Dose\ (mJ/cm^2)}{Lamp\ Intensity\ (mW/cm^2)}$
 - If your lamp intensity is $8.7\ \frac{mJ}{cm^2}$, your exposure time would be
$$time = \frac{110\ mJ/cm^2}{8.7\ mW/cm^2} = 12.6\ seconds$$

Exposure

1. Once you have your exposure time, make sure you turn the dial to correspond with your exposure time
2. Load your wafer into the mask aligner
3. Make sure your wafer is centered in the chuck
4. Bring your wafer in contact with the mask and press the green “Exposure” button
5. After exposure, unload your wafer and bring it to the solvent fume hood

Development

1. Fill a small glass dish with about 50 mL of **AZ 300 MIF Developer**
2. Place your exposed wafer into the developer for **75 seconds**
 - Make sure to agitate the wafer by stirring the glass dish
3. Rinse the wafer using the DI water faucet for a couple of seconds
 - Make sure to grab a different end of the wafer with the tweezers while rinsing
 - There can be developer underneath the tweezers
4. Dry the wafer using the N₂ gun
 - Make sure to dry underneath the tweezers

Inspection

1. Use the microscope to inspect the patterns and test structures on your wafer
 - Look at the resolution structures and record what features survived

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room

- (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
- (c) Wait for the vacuum gauge to reach 300 mtorr
- (d) Turn on the power strip between the gas flow controller and the RIE tool
- (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
- (f) Turn on the main flow and the flow for both O₂ and Ar
- (g) Press the gas flow 1 button on the RIE
- (h) Wait for the vacuum gauge to stabilize
- (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W
- (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**
- (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
- (l) Press the gas flow 1 button on the RIE to turn off the gas flow
- (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
- (n) Turn off the roughening pump underneath the table
- (o) Press the vent button and wait for the chamber to return to atmosphere
- (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
- (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Optional Step

1. Place the wafer back on the hotplate for 2 minutes
 - This is the “Hard Bake”
 - This will strengthen the resist to survive etch chemistry in later steps

Buffered Oxide Etchant (BOE) Etch

Hydrofluoric Acid (HF) Safety

- BOE contains **HF** and it can be **extremely dangerous**
- An exposure area equivalent to the area of your hand can be fatal
 - Attacks calcium in your body
 - Bones and neurons contain calcium
- Make sure your gloves and apron stay dry at all times
- If you see any clear liquid on your PPE, make sure to rinse and dry it before continuing
- Any spills should be reported to your TA
- Make sure to know the location of the HF response kit before proceeding.

Prepare Acid/Base Fume Hood

1. Locate the following items and place them in the acid/base fume hood
 - (a) Two large **teflon** beakers
 - (b) Two small **teflon** beakers
 - (c) One 2in wafer holder
 - This will be made from an old bottle
2. Fill the large teflon beakers with DI water and place them in the back of the fumehood
3. Determine the etch time
 - The etch rate of thermal oxide in BOE is about **2 nm/sec (120 nm/min)**
 - Divide the thickness of your oxide by 100 to determine your etch time
 - Example: 400 nm of oxide will result in a **3-minute, 33-second etch**
 - It is also a good practice to etch for a few seconds longer to make sure all the oxide is gone in the desired areas

Put on PPE

1. You should already have on the standard lab PPE

- Lab coat
- safety glasses
- Gloves
- long pants
- closed-toe shoes

2. Put on the acid/base PPE

- Blue acid apron
- faceshield
- Black chemical gloves

BOE Etch

1. Once PPE is on, locate the BOE waste bottle and verify it is not full
2. Place it in the fume hood
3. Take out the fresh BOE and pour it into one of the small teflon beakers
4. Put away fresh BOE when finished
5. Place the wafer into the wafer holder
6. Place wafer into the BOE etchant for the previously determined time
7. Once wafers have finished etching, take the wafer out of the BOE etchant
 - make sure to allow any etchant on the wafer and in the holder to fall back into the BOE without splashing
8. Place the wafer and holder into the first large beaker of DI water for a few seconds
 - Try to prevent water from splashing or dripping onto the edge of the beaker or inside the fumehood
9. Remove the first beaker and place the wafer into the second beaker of DI water
10. Remove the wafer from the second beaker and place it into the empty small teflon beaker

- Make sure any excess water falls back into the water beaker
11. Take the wafer to the DI water faucet and triple rinse the wafer by filling the beaker with water and emptying it.
 12. Once finished, place the wafer to the side and rinse and dry your gloves
 13. Use the N₂ gun to dry your wafer
 14. Use the filmetrics tool to verify the oxide has been completely etched away
 15. Put the wafer back into the BOE if oxide still remains
 16. Once finished etching, clean up the acid fume hood
 - Pour BOE into the waste container
 - Pour rinse water into the acid/base sink
 - Use the tap water faucet to rinse out all the beakers and wafer holders used
 - Put rinsed tools on the drying rack.
 - Rinse and dry the gloves and put away acid/base PPE

Resist Strip

1. Strip resist using acetone or MLO-07 resist strip
2. Using acetone:
 - (a) Pour acetone into a small glass bowl
 - (b) Place wafer into the beaker and agitate until all the resist has been removed
 - (c) Remove wafer and rinse using DI water faucet and dry using N₂ gun
 - (d) Pour any leftover acetone into the acetone waste container
 - (e) Rinse any dishes used and put them on the drying rack
3. Using MLO-07 resist strip:
 - (a) Pour the MLO-07 into a small bowl and place it on the hot plate in the solvent fume hood
 - This hotplate does **NOT** have a wafer on it
 - (b) Set the hotplate to **75 °C**
 - (c) Use a thermometer to verify the temperature of the solution

- (d) Once the temperature has reached at least 60 °C, place the wafers into the solution and agitate until all the resist has been removed
- (e) Turn off the hotplate and place the glass bowl inside the fume hood
- (f) Remove the wafer and rinse using the DI water faucet and dry with the N₂ gun
- (g) Pour any leftover MLO-07 into the MLO-07 waste container
- (h) Rinse any dishes used and put them on the drying rack

Lab 5: Thermal Oxidation (Gate Oxide)

Prepare Tube Furnace

1. Make sure the oxidation tube is installed in the tube furnace
2. Open both valves on the Oxygen tank
3. Make sure the gas flow valve is set to “Dry Oxidation”
4. Set the oxygen flow for **1 L/min**
5. Turn on the furnace and set it to **1000 °C**
6. You can do the next few steps while the furnace is heating up

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room
 - (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
 - (c) Wait for the vacuum gauge to reach 300 mtorr
 - (d) Turn on the power strip between the gas flow controller and the RIE tool
 - (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
 - (f) Turn on the main flow and the flow for both O₂ and Ar
 - (g) Press the gas flow 1 button on the RIE
 - (h) Wait for the vacuum gauge to stabilize
 - (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W
 - (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**

- (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
- (l) Press the gas flow 1 button on the RIE to turn off the gas flow
- (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
- (n) Turn off the roughening pump underneath the table
- (o) Press the vent button and wait for the chamber to return to atmosphere
- (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
- (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Thermal Dry Oxidation

1. Place the wafers into the quartz boat meant for thermal oxidation
2. Once all the wafers have been loaded, place a wafer on the front and back ends of the boat
 - These wafers should sandwich your process wafers
3. **PUT ON THE HEAT-RESISTANT GLOVES BEFORE PROCEEDING**
4. Remove the glass end cap from the tube furnace
5. Using the metal plate and the tongs, place the boat holding the wafers into the turn furnace
6. Once the wafers are placed, use the quartz rod to push the wafers into the center of the tube
 - **MAKE SURE YOU DO THIS VERY SLOWLY OR THE WAFERS AND/OR BOAT WILL CRACK DUE TO THERMAL STRESS**
 - You should move about 1 in every 10 seconds
 - There should be a Sharpie mark on the quartz rod
 - The wafers will be centered in the tube once this mark lines up with the edge of the tube
 - Take care to not bang the rod against the furnace, as this could cause the rod to shatter

- **THE END OF THE ROD WILL REACH THE SAME TEMPERATURE AS THE FURNACE, SO MAKE SURE TO KEEP IT AWAY FROM YOUR BODY AND OTHER PEOPLE**

7. Once the wafers have been centered in the tube, remove the quartz rod
8. Let the wafers diffuse for **45 minutes**
9. After 45 minutes, remove the wafers from the tube furnace following all rules described during loading
10. Remove the wafers from the boat and place them into your acrylic box

- **THE BOAT WILL STILL BE EXTREMELY HOT, SO BE CAREFUL WHEN RETRIEVING THE WAFERS**

11. Use the Filmmetrics tool to measure the oxide thickness in the gate region

Lab 6: Mask #3 Lithography: Contact Regions and BOE Etch

Lithography

Prepare the Wafers

1. Turn on the photolithography hotplate
2. Wait for the hotplate to reach **112 °C**
 - The hotplate will overshoot before stabilizing at 112 °C
3. Place the wafers on the hot plate and let them bake for **10 minutes**
 - This is the dehydration bake

Prepare the Spinner

1. While the wafers are dehydrating, set up the spinner — using the **Lithography SOP**
 - (a) Fold a piece of foil into a square and tear off the corner where the folds meet
 - (b) Unfold the foil and fold it onto the spinner.
 - (c) located the 2 in-chuck and place it onto the spinner
 - (d) turn on the vacuum pump located underneath the left cabinet, underneath the spinner area
 - (e) turn on the spinner located to the left of the hotplate
2. **MAKE SURE THE OVERHEAD LIGHTS ARE TURNED OFF BEFORE PROCEEDING TO THE NEXT STEP**

Photoresist Spinning

1. Once the wafers have finished dehydrating, locate the bottle of **AZ 1512** photoresist and place it next to the spinner
 - This is a **positive** photoresist
2. Center your wafer on the chuck and set your spin parameters on the spinner
 - Speed: **1800 RPM**
 - Time: **30 seconds**

3. use the nitrogen blower to blow off any large particles or dust from the surface of your wafer
4. Dispense about 1.5 mL of photoresist onto the center of your wafer, then press the foot pedal to start spinning
 - This does not need to be exact
5. Once the spinner has stopped, place your wafer onto the hotplate for **2 minutes**
6. Remove the wafer from the hotplate and put it back into the acrylic case

Prepare the Mask Aligner

1. Turn on the light located above the mask aligner
2. turn on the power strip located to the right of the mask aligner
3. Turn on the vacuum pump located on the floor to the right of the mask aligner
4. Make sure the mask aligner is turned on
5. locate **Mask #3: Contact Areas** in the TA drawer
6. Place the mask onto the mask holder and turn on the vacuum switch located on the mask aligner
 - (a) There is a specific orientation the mask should be in before placing it onto the holder
 - (b) The bottom of the pattern should be facing up
 - (c) Ask your TA to show you how to identify the correct orientation
7. Verify the mask is secured by flipping it upside down with your hand underneath it
8. Once you confirm the mask is secure, load the mask into the mask aligner
9. Next, perform a test exposure to find the lamp intensity
10. Calculate the exposure time using the current lamp intensity and a dose of **110 $\frac{mJ}{cm^2}$**
 - The equation for exposure time is $time\ (seconds) = \frac{Dose\ (mJ/cm^2)}{Lamp\ Intensity\ (mW/cm^2)}$
 - If your lamp intensity is $8.7\ \frac{mJ}{cm^2}$, your exposure time would be
$$time = \frac{110\ mJ/cm^2}{8.7\ mW/cm^2} = 12.6\ seconds$$

Exposure

1. Once you have your exposure time, make sure you turn the dial to correspond with your exposure time
2. Load your wafer into the mask aligner
3. Make sure your wafer is centered in the chuck
4. Bring your wafer in contact with the mask and press the green “Exposure” button
5. After exposure, unload your wafer and bring it to the solvent fume hood

Development

1. Fill a small glass dish with about 50 mL of **AZ 300 MIF Developer**
2. Place your exposed wafer into the developer for **75 seconds**
 - Make sure to agitate the wafer by stirring the glass dish
3. Rinse the wafer using the DI water faucet for a couple of seconds
 - Make sure to grab a different end of the wafer with the tweezers while rinsing
 - There can be developer underneath the tweezers
4. Dry the wafer using the N₂ gun
 - Make sure to dry underneath the tweezers

Inspection

1. Use the microscope to inspect the patterns and test structures on your wafer
 - Look at the resolution structures and record what features survived

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room

- (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
- (c) Wait for the vacuum gauge to reach 300 mtorr
- (d) Turn on the power strip between the gas flow controller and the RIE tool
- (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
- (f) Turn on the main flow and the flow for both O₂ and Ar
- (g) Press the gas flow 1 button on the RIE
- (h) Wait for the vacuum gauge to stabilize
- (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W
- (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**
- (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
- (l) Press the gas flow 1 button on the RIE to turn off the gas flow
- (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
- (n) Turn off the roughening pump underneath the table
- (o) Press the vent button and wait for the chamber to return to atmosphere
- (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
- (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Optional Step

1. Place the wafer back on the hotplate for 2 minutes
 - This is the “Hard Bake”
 - This will strengthen the resist to survive etch chemistry in later steps

Buffered Oxide Etchant (BOE) Etch

Hydrofluoric Acid (HF) Safety

- BOE contains **HF** and it can be **extremely dangerous**
- An exposure area equivalent to the area of your hand can be fatal
 - Attacks calcium in your body
 - Bones and neurons contain calcium
- Make sure your gloves and apron stay dry at all times
- If you see any clear liquid on your PPE, make sure to rinse and dry it before continuing
- Any spills should be reported to your TA
- Make sure to know the location of the HF response kit before proceeding.

Prepare Acid/Base Fume Hood

1. Locate the following items and place them in the acid/base fume hood
 - (a) Two large **teflon** beakers
 - (b) Two small **teflon** beakers
 - (c) One 2in wafer holder
 - This will be made from an old bottle
2. Fill the large teflon beakers with DI water and place them in the back of the fumehood
3. Determine the etch time
 - The etch rate of thermal oxide in BOE is about **2 nm/sec (120 nm/min)**
 - Divide the thickness of your oxide by 100 to determine your etch time
 - Example: 400 nm of oxide will result in a **3-minute, 33-second etch**
 - It is also a good practice to etch for a few seconds longer to make sure all the oxide is gone in the desired areas

Put on PPE

1. You should already have on the standard lab PPE

- Lab coat
- safety glasses
- Gloves
- long pants
- closed-toe shoes

2. Put on the acid/base PPE

- Blue acid apron
- faceshield
- Black chemical gloves

BOE Etch

1. Once PPE is on, locate the BOE waste bottle and verify it is not full
2. Place it in the fume hood
3. Take out the fresh BOE and pour it into one of the small teflon beakers
4. Put away fresh BOE when finished
5. Place the wafer into the wafer holder
6. Place wafer into the BOE etchant for the previously determined time
7. Once wafers have finished etching, take the wafer out of the BOE etchant
 - make sure to allow any etchant on the wafer and in the holder to fall back into the BOE without splashing
8. Place the wafer and holder into the first large beaker of DI water for a few seconds
 - Try to prevent water from splashing or dripping onto the edge of the beaker or inside the fumehood
9. Remove the first beaker and place the wafer into the second beaker of DI water
10. Remove the wafer from the second beaker and place it into the empty small teflon beaker

- Make sure any excess water falls back into the water beaker
11. Take the wafer to the DI water faucet and triple rinse the wafer by filling the beaker with water and emptying it.
 12. Once finished, place the wafer to the side and rinse and dry your gloves
 13. Use the N₂ gun to dry your wafer
 14. Use the filmetrics tool to verify the oxide has been completely etched away
 15. Put the wafer back into the BOE if oxide still remains
 16. Once finished etching, clean up the acid fume hood
 - Pour BOE into the waste container
 - Pour rinse water into the acid/base sink
 - Use the tap water faucet to rinse out all the beakers and wafer holders used
 - Put rinsed tools on the drying rack.
 - Rinse and dry the gloves and put away acid/base PPE

Resist Strip

1. Strip resist using acetone or MLO-07 resist strip
2. Using acetone:
 - (a) Pour acetone into a small glass bowl
 - (b) Place wafer into the beaker and agitate until all the resist has been removed
 - (c) Remove wafer and rinse using DI water faucet and dry using N₂ gun
 - (d) Pour any leftover acetone into the acetone waste container
 - (e) Rinse any dishes used and put them on the drying rack
3. Using MLO-07 resist strip:
 - (a) Pour the MLO-07 into a small bowl and place it on the hot plate in the solvent fume hood
 - This hotplate does **NOT** have a wafer on it
 - (b) Set the hotplate to **75 °C**
 - (c) Use a thermometer to verify the temperature of the solution

- (d) Once the temperature has reached at least 60 °C, place the wafers into the solution and agitate until all the resist has been removed
- (e) Turn off the hotplate and place the glass bowl inside the fume hood
- (f) Remove the wafer and rinse using the DI water faucet and dry with the N₂ gun
- (g) Pour any leftover MLO-07 into the MLO-07 waste container
- (h) Rinse any dishes used and put them on the drying rack

Lab 7: Mask #4 Lithography: Metal and Metal Sputtering

Lithography

Prepare the Wafers

1. Turn on the photolithography hotplate
2. Wait for the hotplate to reach **112 °C**
 - The hotplate will overshoot before stabilizing at 112 °C
3. Place the wafers on the hot plate and let them bake for **10 minutes**
 - This is the dehydration bake

Prepare the Spinner

1. While the wafers are dehydrating, set up the spinner — using the **Lithography SOP**
 - (a) Fold a piece of foil into a square and tear off the corner where the folds meet
 - (b) Unfold the foil and fold it onto the spinner.
 - (c) located the 2 in-chuck and place it onto the spinner
 - (d) turn on the vacuum pump located underneath the left cabinet, underneath the spinner area
 - (e) turn on the spinner located to the left of the hotplate
2. **MAKE SURE THE OVERHEAD LIGHTS ARE TURNED OFF BEFORE PROCEEDING TO THE NEXT STEP**

Photoresist Spinning

1. Once the wafers have finished dehydrating, locate the bottle of **nLOF 2035** photoresist and place it next to the spinner
 - This is a **Negative** photoresist
2. Center your wafer on the chuck and set your spin parameters on the spinner
 - Speed: **3500 RPM**
 - Time: **30 seconds**

3. use the nitrogen blower to blow off any large particles or dust from the surface of your wafer
4. Dispense about 1.5 mL of photoresist onto the center of your wafer, then press the foot pedal to start spinning
 - This does not need to be exact
5. Once the spinner has stopped, place your wafer onto the hotplate for **3 minutes**
6. Remove the wafer from the hotplate and put it back into the acrylic case

Prepare the Mask Aligner

1. Turn on the light located above the mask aligner
2. turn on the power strip located to the right of the mask aligner
3. Turn on the vacuum pump located on the floor to the right of the mask aligner
4. Make sure the mask aligner is turned on
5. locate **Mask #2: Gate Oxide** in the TA drawer
6. Place the mask onto the mask holder and turn on the vacuum switch located on the mask aligner
 - (a) There is a specific orientation the mask should be in before placing it onto the holder
 - (b) The bottom of the pattern should be facing up
 - (c) Ask your TA to show you how to identify the correct orientation
7. Verify the mask is secured by flipping it upside down with your hand underneath it
8. Once you confirm the mask is secure, load the mask into the mask aligner
9. Next, perform a test exposure to find the lamp intensity
10. Calculate the exposure time using the current lamp intensity and a dose of **110 $\frac{mJ}{cm^2}$**
 - The equation for exposure time is $time\ (seconds) = \frac{Dose\ (mJ/cm^2)}{Lamp\ Intensity\ (mW/cm^2)}$
 - If your lamp intensity is $8.7\ \frac{mJ}{cm^2}$, your exposure time would be
$$time = \frac{110\ mJ/cm^2}{8.7\ mW/cm^2} = 12.6\ seconds$$

Exposure

1. Once you have your exposure time, make sure you turn the dial to correspond with your exposure time
2. Load your wafer into the mask aligner
3. Make sure your wafer is centered in the chuck
4. Bring your wafer in contact with the mask and press the green “Exposure” button
5. **AFTER EXPOSURE, UNLOAD YOUR WAFER AND PLACE IT ON THE HOT PLATE FOR 2 MINUTES**
 - This is the post-exposure bake

Development

1. Fill a small glass dish with about 50 mL of **AZ 300 MIF Developer**
2. Place your exposed wafer into the developer for **2 minutes**
 - Make sure to agitate the wafer by stirring the glass dish
3. Rinse the wafer using the DI water faucet for a couple of seconds
 - Make sure to grab a different end of the wafer with the tweezers while rinsing
 - There can be developer underneath the tweezers
4. Dry the wafer using the N₂ gun
 - Make sure to dry underneath the tweezers

Inspection

1. Use the microscope to inspect the patterns and test structures on your wafer
 - Look at the resolution structures and record what features survived

Descum

1. Take wafers to the table top RIE (Reactive Ion Etching) tool
2. Before descuming wafers, warm up the RIE tool
 - (a) Open the valves on the Oxygen and Argon green cylinders in the corner of the room
 - (b) Make sure the top lid is closed and turn on the roughening pump underneath the table
 - Hold down the lid during the first few seconds of the pump down
 - You can let go once the vacuum level starts to decrease
 - (c) Wait for the vacuum gauge to reach 300 mtorr
 - (d) Turn on the power strip between the gas flow controller and the RIE tool
 - (e) On the gas flow controller, set **O₂ to 0.09 sccm** and **Ar to 0.4 sccm**
 - (f) Turn on the main flow and the flow for both O₂ and Ar
 - (g) Press the gas flow 1 button on the RIE
 - (h) Wait for the vacuum gauge to stabilize
 - (i) Press the RF power button and slowly bring the power up to 90 W
 - Reflected power should stay below 5 W
 - If it is 5 or above, slowly lower the power until it goes under 5 W
 - (j) Once the power reaches **90 W**, let the RIE stay there for **30 seconds**
 - (k) After 30 seconds, slowly lower the power to 0 W, then press the RF power button to turn off the power
 - (l) Press the gas flow 1 button on the RIE to turn off the gas flow
 - (m) Turn off the main gas flow and the gas flow for O₂ and Ar on the gas flow controller
 - (n) Turn off the roughening pump underneath the table
 - (o) Press the vent button and wait for the chamber to return to atmosphere
 - (p) Load the wafers and repeat steps (b) through (o) to descum the wafers
 - (q) Unload the wafers and make sure the gas flow controller is turned off, and the valves on the green gas cylinders are completely closed

Metal Sputtering

Prepare The Sputter Tool

1. Make sure the AlSi target is installed in the sputter gun
 - There should be a tag that displays what target is installed
2. Vent the chamber using the staff **Sputter SOP**
3. Take the chucks out of the chamber and load your wafers onto them
4. Once your wafers are loaded, install the chucks back into the chamber
5. Set the rotating platform to 30% and close the chamber door
6. Pump down the chamber using the staff **Sputter SOP**
7. Once the chamber is pumped down, use the staff **Sputter SOP** to run the sputter for **30 minutes**
 - (a) This should be about 300 nm of metal
8. After 30 minutes, use the staff **Sputter SOP** to vent the chamber and unload the wafers
9. Make sure the chamber is pumped down using the staff sputter SOP

Lab 8: Metal Liftoff and Thermal Anneal

Prepare the Liftoff Bath

1. Pour some MLO-07 into a small bowl and place it on the hot plate in the solvent fume hood
 - This hotplate does **NOT** have a wafer on it
2. Set the hotplate to **75 °C**
3. Use a thermometer to verify the temperature of the solution

Prepare the Tube Furnace

1. Make sure the diffusion tube is installed in the furnace
2. Make sure the SOG/Diffusion tube is installed in the tube furnace
3. Make sure the gas flow valve is set to “Dry Oxidation”
 - We are not flowing oxygen during this step, so there should be no oxidation
 - The tube furnace should be connected to the house nitrogen line
4. Set the nitrogen flow for **1 L/min**
5. Turn on the furnace and set it to **550 °C**
6. You can do the next few steps while the furnace is heating up

Prepare the Sonicator

1. Remove the lid from the sonicator
2. Make sure the sonicator is filled with DI water
3. Verify the sonicator is plugged into the wall

Liftoff

1. Verify the temperature of the bath is at least 60 °C
2. Place the wafers into the bath for a few minutes
 - You should immediately start to see metal floating off of the wafer
3. After a few minutes, take the glass bowl off the hot plate and place it into the sonicator bath
 - The bowl should not be submerged completely in the water bath
 - You should be holding the top edge of the bowl while the bottom 3rd of the bowl should be in the water
4. Hold the bowl there for a few seconds, then place the bowl back onto the hot plate
5. All the metal should be removed from the undesired areas
6. If not, repeat the above steps as necessary
7. Once all the metal has been removed from the undesired areas, rinse the wafer with DI water and dry using the N₂ gun
8. Use the profilometer to measure the thickness of the metal

Thermal Anneal

1. Before doing thermal anneal, use the probe station to measure the contact resistance of your devices
2. Place the wafers onto the quartz boat used in the diffusion lab
3. Slide the wafers into the center of the tube and let them anneal for **5 minutes**
 - You do not need to go slow for this step since the temperature is much lower
4. Remove your wafers from the furnace and measure the contact resistance of your devices again
 - The contact resistance should have decreased
5. Congratulations! You have finished the fabrication of your devices!